Dialogue Nine: Derivative

Author: The previous dialogue gave us an opportunity to introduce the concept of a derivative for a specific example from physics (the instantaneous velocity of a body moving nonuniformly along a straight line). Now let us examine this concept from a purely mathematical viewpoint without assigning any physical meaning to the mathematical symbols used.

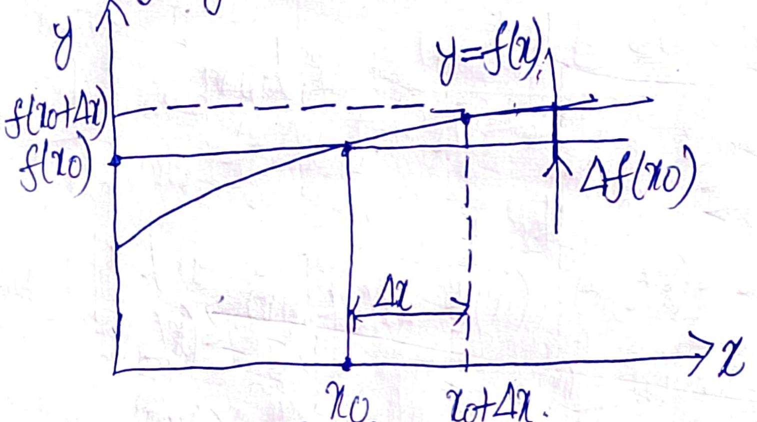

Fig. 36

Figure 36 shows a graph of an arbitrary function $y=f(x)$. Let us select an arbitrary point $x=x_0$ from the domain of the function. In the subsequent argument this point is considered as fixed. Now consider another point x from the domain of the function and introduce a notation $\Delta x = x - x_0$. The value Δx is called the increment of the independent variable. The increment is considered with respect to the fixed point x_0. Depending on the point x, the value of Δx may be larger or smaller, positive or negative.

Now let us examine a difference between the values of the function at points $x = x_0 + \Delta x$ and $x = x_0$: $\Delta f(x_0) = f(x_0+\Delta x) - f(x_0)$. The increment $\Delta f(x_0)$ is said to be the increment of a function f at a point x_0. Since x_0 is fixed, $\Delta f(x_0)$ should be considered as a function of a variable increment Δx of the independent variable.

Reader: Then it is probably more logical to denote this function by $\Delta f(\Delta x)$, and not by $\Delta f(x_0)$, isn't it?